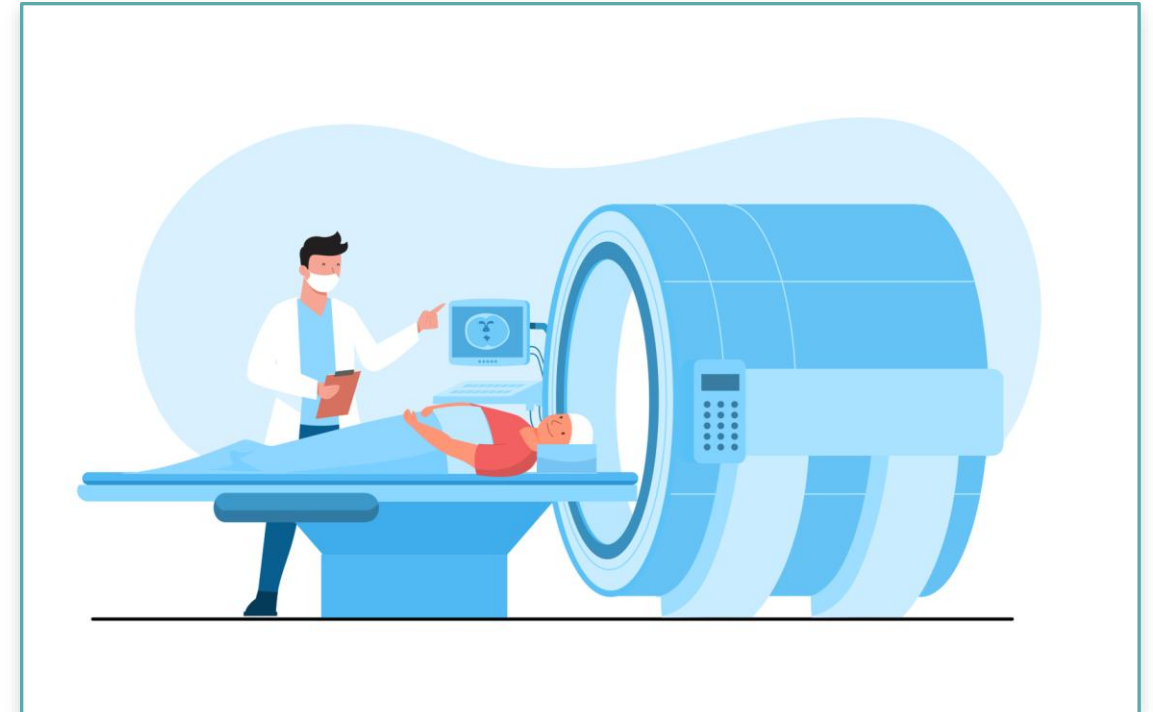


RADIOLOGY MEDICAL EDUCATION • CASE STUDY

Never-Skilling

Meet Dr. R

- Dr. R is a PGY-2 diagnostic radiology resident at a large academic center.
- Her institution has an AI musculoskeletal detection tool on every hip and pelvis radiograph that flags findings, such as early-stage avascular necrosis (AVN) of the femoral head.
- For over a year, nearly every hip film Dr. R has read has had the AI flag AVN on each image for her.



An Example



AP pelvis — AI-highlighted subchondral lucency (“crescent sign”), left femoral head

<https://radiopaedia.org/cases/osteonecrosis-of-the-femoral-head-6?lang=us>

- A 34-year-old on chronic steroids presents with vague hip pain and Dr. R orders an AP pelvis film.
- The AI highlights a “crescent sign” in the superolateral femoral head, representing subchondral lucency consistent with early AVN.
- AVN is important to catch early, as treatment is most effective before femoral head collapse and joint destruction.

Alone With the Film



AP pelvis — subtle bilateral early AVN, no AI support, no annotation

- On a community away rotation without AI on the worklist, a similar patient with hip pain presents to the ED.
- Dr. R. reviews the film but reports it as unremarkable.
- When the attending flags the finding and asks her to explain her reasoning, she struggles because she never developed the underlying diagnostic skill. This is an example of never-skilling.

Never-skilling:

The failure to develop foundational clinical reasoning competencies during formative training due to excessive AI substitution for cognitive effort.